Peter A. Lilley

Software Engineer - Local LLM Inference, Language-to-Geometry Systems, Production Tooling

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SUMMARY

I am a software engineer. I build production systems that other professionals use daily. I built a language-to-geometry pipeline. It uses a plain-language interface to generate 3D models and set up engineering deliverables. I use locally hosted LLM inference to handle intent. I use deterministic C# and .NET code to construct and validate geometry. I also build OCR ingestion and automation tools. These tools cut repetitive drafting from 8 to 12 hours to about 30 seconds per cycle. I work in C#, Python, C++17, TypeScript, and Dart.

SKILLS

Languages: C#, Python, C++, Dart, TypeScript, JavaScript, SQL, LISP, VBA

AI and local models: Ollama, LM Studio, Open WebUI, Google APIs, OCR, speech-to-text, prompt engineering, text matching

Databases and Data Systems: SQLite, Firebase, Postgres, XData schemas, data modeling, JSON structuring, graph calculators

Engineering and math: Matrix math, 3D graphics, Express, React, Git, CMake

Domain systems and quality: CAD plugins, Revit automation, 3D metadata, ARCore, part dimensioning, coordinate measuring machines, ISO 9001

EXPERIENCE

David Mason & Associates - Staff Technician (CAD automation / Civil 3D tools) Jul 2024 - Present

St. Louis, MO

- I build CAD generation tools. These tools turn natural-language input into 3D models and drawing setups.
- I use local LLM inference for intent. I use deterministic C# and .NET for geometry and validation.
- I introduced this capability at the firm.
- I built OCR ingestion software. It parses existing PDF plan sets to auto-check and auto-populate them. This replaced a manual mark-up loop.
- I automate repetitive drafting with Python and Dynamo. This cuts cycles from 8 to 12 hours to about 30 seconds. It reduces errors before licensed-engineer review by about 25%.
- I design typed metadata schemas. These schemas attach to model entities. I coordinate standards automation across civil, mechanical, plumbing, and architectural disciplines.
- I built C# plugins for Civil 3D to automate layout and title block renaming. This tool parses hundreds of text entities to enforce company standards.
- It executes in under 1.7 seconds from a few text prompts. This eliminates 15-to-20-second manual loading delays between clicks.
- I also wrote commands to audit drawings, fix broken data, and export logs showing exactly which layers and objects changed.

Component Bar Products - Quality Engineer - Manufacturing / Precision Machining Apr 2023 - Jun 2024
O'Fallon, MO

- I programmed and operated coordinate measuring machines. I verified precision-machined parts to +/-0.001 inch or tighter against GD&T specifications.
- I validated dimensional tolerances against CAD models and customer prints. Machining problems showed as geometry mismatches, not downstream scrap.
- I enforced ISO 9001 across production lines. I integrated Production Part Approval Process documentation. I managed calibration and traceability of metrology assets.

Heideman & Associates, Inc. - Revit/CAD Technician - Mechanical/Plumbing May 2018 - Apr 2019
Fenton, MO

- I produced mechanical and plumbing construction documents. I built as-built models in Revit and AutoCAD.
- I audited standards for cross-discipline consistency.
- I built automated data libraries for building-system objects. I standardized hundreds of outdated details.

PROJECTS

Digital Twin Pro

Personal product, Free Parameter LLC | Flutter/Dart 3, SQLite, Firebase, Google Gemini API, ARCore | github.com/freeParameterized/digital-twin-pro

- I designed, funded, and shipped a cross-platform 3D inventory application. I released it to a Google Play beta and built a Windows desktop version.
- I wrote the 3D renderer by hand instead of using a game engine. I used a painter's-algorithm scene rendering with a Z-sorted draw queue and custom projection.
- I added photo-based item detection and voice entry via a paid Gemini API.
- I built the database with SQLite to store inventory data locally. I included a move audit log. I added QR booklets and CSV/JSON export.

CAD integration bridge

C++17, Dear ImGui, Windows COM | local repository

- I built a C++17 Dear ImGui host. It COM-bridges BricsCAD, AutoCAD, and Civil 3D.
- It discovers installed SDKs at runtime. It hot-loads LISP, C#, and BRX plugins. It has about 14,000 lines of code.

Interactive Portfolio (this site)

React, TypeScript, Three.js, Express, Ollama

- I built a React Three Fiber portfolio over an Express API.
- It retrieves data from a curated corpus. It answers questions through a local model.
- It uses on-device speech. It degrades to extractive answers when no model runs.

EDUCATION

Calculus II (CLEP Credit) - Saint Louis University (SLU) 2018
St. Louis, MO

Coursework toward Latin American Studies with Technical Applications - Missouri University of Science and Technology 2019 - 2020
Rolla, MO

I completed rigorous, no-calculator College Algebra. This included determinants, Cramer's rule, and introductory linear algebra concepts. I also took general education courses. I have technical fluency in Spanish.

A.A.S., Building Systems Engineering Technology - Ranken Technical College 2019
Wentzville, MO